Raw sequence data

Data formats

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Learning outcomes

- **Understand data formats used in NGS**
  - FASTQ
  - FASTA
  - fast5
- **Encoded quality scores**
  - PHRED scores

FASTA format

Nucleotide or AA sequences

- Always starts with “>” followed by an identifier. The rest of the line (a description) is optional.
- Nucleotide or animo acid sequence follows on subsequent lines
- This format is used for reference genomes, and reference-based alignment workflows

```
[jlrhodes@login-a live]$ head -5 GCF_000002655.1_ASM265v1_genomic.fa
>NC_007194.1 Aspergillus fumigatus Af293 chromosome 1, whole genome shotgun sequence
cctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccc
taaccctaaccctaaccctaaccctaaccctaaccctaaccttTAGGCAACTGCAGCTTCAAACCAGATTTGGATGGGC
CACACGCGTGCTAGGTTTCCTGGTTCTTGGAACGACATTGTTCTCACCTAGTGTGATGAGCGTTCGTCAAGTTCCAAAGC
AAAGACGCACTTTGATCGAGTTTTCAGCGTTCAAGGAGGTCCCGTGTATGCTGTTCTGCGTAGCGATGTTCTTTGGTTAT
[jlrhodes@login-a live]$
```

FASTA format

References

- Reference genomes can be found on GenBank, and new versions are released periodically.
- Reference genomes used for alignment workflows must be indexed for use by three separate pieces of software: BWA, Samtools and GATK
  - indexing allows the aligner software to narrow down the potential origin of a query sequence within the genome, saving time and computational memory

```
[jlrhodes@login-a live]$ head -5 GCF_000002655.1_ASM265v1_genomic.fa
>NC_007194.1 Aspergillus fumigatus Af293 chromosome 1, whole genome shotgun sequence
cctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccctaaccc
taaccctaaccctaaccctaaccctaaccctaaccctaaccttTAGGCAACTGCAGCTTCAAACCAGATTTGGATGGGC
CACACGCGTGCTAGGTTTCCTGGTTCTTGGAACGACATTGTTCTCACCTAGTGTGATGAGCGTTCGTCAAGTTCCAAAGC
AAAGACGCACTTTGATCGAGTTTTCAGCGTTCAAGGAGGTCCCGTGTATGCTGTTCTGCGTAGCGATGTTCTTTGGTTAT
[jlrhodes@login-a live]$
```

FASTA format

Indexing a reference genome for alignment

- Index using BWA
  - `bwa index reference.fa`
  - Type ‘ls -lh’ to see the files created
- Index using Samtools
  - `samtools faidx reference.fa`
  - Creates a file with extension ‘.fai’, which contains one record per line for each of the costings (contiguous DNA segments) in the fasta file
- Index for GATK using Picard
  - Creates a ‘dictionary’ (.dict file) using Picard in order to call SNPs using GATK. This .dict file describes the contents of your reference fasta file
  - `picard CreateSequenceDictionary -R reference.fa -O reference.dict`

FASTQ format

Raw sequence data

- FASTQ files are compressed with ‘gzip’, and have the .gz extension to save space. We use ‘gunzip’ to decompress and look at the contents, but all software will use compressed fastq files, so there is no need to decompress

```
[jlrhodes@login-c Af293_1]$ head -4 170309_K00166_0192_BHHMH3BBXX_6_TP-D7-012_TP-D5-006_1.fastq
@K00166:192:HHMH3BBXX:6:1101:4807:1068 1:N:0:NGCGATAG+NAAGATTA
NTTGTAGCGACTGGTGAAGAAGTCTGTGGTTTTGGTTGCCAGGAAGCTGTACTCAATAATGGACTTAGTTCAGTGTCCACACTTCCCCTTGTGGCTCTCGACGATGGCGAGGAGCATGATGGGGAATGGATGAAAGAG
GCGGAAGTTGAC
+
#AAAFJJJJJJJJJJJJJJJJJJJJJJJJJFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJFJJJJJJAJJJJJJJ<JJJJJJJJJJJJFAJJJJJJJJFJJ<FFJAJJFJ-A-AJ--7FJF<FA<FAF<-7<AFAAF
-)-7FJJJ--<F
```

FASTQ format

Raw sequence data

- FASTQ files contain the sequence data and associated quality data over four lines
- Line 1: here, ‘K00166’ is the name of the sequencing machine used. The rest of the data on this line is optional.
    - This line is in ‘Casava 1.8’ format, which is used in Illumina 1.9 and beyond.
    - ‘192’ is the run ID, HHMH3BBXX is the flow cell ID; ‘6’ is the flow cell lane; ‘1101’ is the tile number within the flow cell lane; 4807 is the ‘x’-coodinate of the cluster within the tile, and 1068 is the ‘y’-coorindate
    - After the space, the ‘1’ means paired end data (it would be ‘2’ for mate pair), ’N’ means reads are not filtered, and the letters at the end are the index sequence
- Line 2 is the raw sequence
- Line 3 begins with ‘+’ and can contain sequence identifiers, but this is optional
- Line 4 is the associated quality data for sequence in line 2 - note there is one quality score per nucleotide

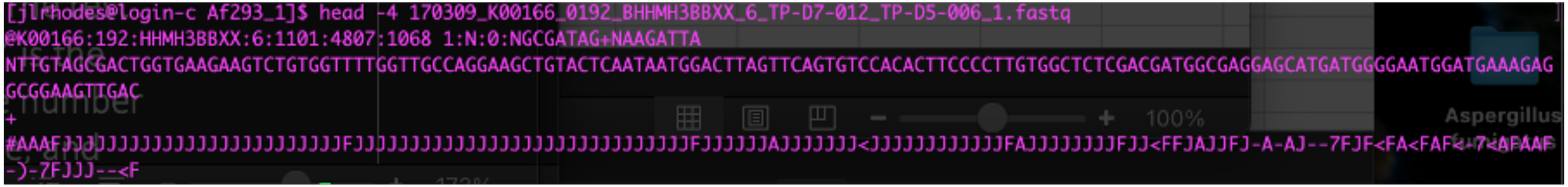
```
[jlrhodes@login-c Af293_1]$ head -4 170309_K00166_0192_BHHMH3BBXX_6_TP-D7-012_TP-D5-006_1.fastq
@K00166:192:HHMH3BBXX:6:1101:4807:1068 1:N:0:NGCGATAG+NAAGATTA
NTTGTAGCGACTGGTGAAGAAGTCTGTGGTTTTGGTTGCCAGGAAGCTGTACTCAATAATGGACTTAGTTCAGTGTCCACACTTCCCCTTGTGGCTCTCGACGATGGCGAGGAGCATGATGGGGAATGGATGAAAGAG
GCGGAAGTTGAC
+
#AAAFJJJJJJJJJJJJJJJJJJJJJJJJJFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJFJJJJJJAJJJJJJJ<JJJJJJJJJJJJFAJJJJJJJJFJJ<FFJAJJFJ-A-AJ--7FJF<FA<FAF<-7<AFAAF
-)-7FJJJ--<F
```

FASTQ format

Quality scores

- Line 4 is the associated quality data for sequence in line 2 - note there is one quality score per nucleotide
- These quality scores are what's known as 'PHRED+33' encoding
  - ! represents the lowest quality
  - J represents the highest quality

```
!"#$%&'()*+,./0123456789:;<=>?@ABCDEFGHIJ
```



ONT long read data format

fast5 format

- Native container for data from ONT platforms, containing raw electrical signal levels measured by the nanopores
  - “squiggle” signals
  - hdf5-based
- Basecalling on fast5 files to acquire FASTQ
  - Do not have sufficient information for methylation calling, so important to keep raw signals
- No formal specification for fast5 format

PacBio long read data format

‘movie’ files

- Raw data from sequel platform comes as BAM, FASTA and FASTQ
- FASTQ same as Illumina files, except that each read will be very long
- HiFi reads contain PHRED scores and use a similar encoding to Illuminate 1.8+